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Common drain for fluid chambers of a motor vehicle drivetrain fluid system

Abstract

A fluid system for a motor vehicle drivetrain is provided including a housing including drain hole, a first fluid chamber for receiving fluid, and a second fluid chamber for receiving fluid, the drain hole coupled to the first fluid chamber and the second fluid chamber for draining fluid from both the first fluid chamber and the second fluid chamber. The fluid system further includes a stopper configured to prevent fluid from draining from the first fluid chamber and the second fluid chamber out of the drain hole in a first orientation of the stopper and configured to allow fluid to drain from the first fluid chamber and the second fluid chamber out of the drain hole in a second orientation of the stopper. The stopper is configured to isolate the first fluid chamber from the second fluid chamber in the first orientation of the stopper.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2001/0047996	12/2000	Weingaertner	220/571	F01M 11/0004
2009/0026014	12/2008	Martin, III	184/1.5	F16H 57/0408
2019/0234260	12/2018	VanDrie	N/A	F01M 11/0408

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Background/Summary

(1) The present disclosure relates generally to motor vehicle drivetrains and more specifically to fluid systems in motor vehicle drivetrains.

BACKGROUND

(2) Motor vehicle drivetrains include fluid chambers to lubricate rotating parts and/or to cool heat generating components.

SUMMARY

(3) A fluid system for a motor vehicle drivetrain is provided including a housing including drain hole, a first fluid chamber for receiving fluid, and a second fluid chamber for receiving fluid, the drain hole coupled to the first fluid chamber and the second fluid chamber for draining fluid from both the first fluid chamber and the second fluid chamber. The fluid system further includes a stopper configured to prevent fluid from draining from the first fluid chamber and the second fluid chamber out of the drain hole in a first orientation of the stopper and configured to allow fluid to drain from the first fluid chamber and the second fluid chamber out of the drain hole in a second orientation of the stopper. The stopper is configured to isolate the first fluid chamber from the second fluid chamber in the first orientation of the stopper.

(4) In examples, the first fluid chamber includes a first drain line emptying into the drain hole and the second fluid chamber includes a second drain line emptying into the drain hole.

(5) In examples, the first drain lines empties into a side of the drain hole and the second drain lines empties into a top of the drain hole.

(6) In examples, a first surface of the stopper blocks an outlet of the first drain line in the first

orientation, and a second surface of the stopper blocks an outlet of the second drain line in the first orientation.

(7) In examples, the stopper includes a seal for sealing the first fluid chamber and the second fluid chamber from an environment outside of the housing.

(8) In examples, the stopper is a drain plug removably installable on the housing to plug the drain hole, a portion of the drain plug being installed in the drain hole in the first orientation, the drain plug being removed from the drain hole in the second orientation.

(9) In examples, the drain plug includes: an exterior section positioned outside of the drain hole in the first orientation; and an interior section positioned inside of the drain hole in the first orientation.

(10) In examples, the first fluid chamber includes a first drain line emptying into the drain hole and the second fluid chamber includes a second drain line emptying into the drain hole, wherein a first surface of the interior section of the drain plug blocks an outlet of the first drain line in the first orientation, and a second surface of the interior section of the drain plug blocks an outlet of the second drain line in the first orientation.

(11) In examples, the interior section of the drain plug includes an external thread and the drain holes includes an internal thread, the external thread being screwed into the internal thread in the first orientation.

(12) In examples, the exterior section of the drain plug includes a head configured to be torqued by a tool to screw the drain plug into the drain hole and into the first orientation, and to unscrew the drain plug from the drain hole and into the second orientation.

(13) In examples, the drain hole has a center axis about which the drain plug is rotatable to screw the drain plug into the drain hole and to unscrew the drain plug from the drain hole, the first fluid chamber including a first drain line emptying radially into the drain hole with respect to the center axis, the second fluid chamber including a second drain line emptying axially into the drain hole with respect to the center axis.

(14) In examples, the drain plug includes a seal for sealing the first fluid chamber and the second fluid chamber from an environment outside of the housing in the first orientation, the seal being sandwiched between the exterior section and the housing in the first orientation.

(15) In examples, the drain hole, the first fluid chamber and the second fluid chamber are configured to cause fluid from the first fluid chamber and fluid from the second fluid chamber to merge inside the drain hole in the second orientation of the stopper.

(16) A motor vehicle drivetrain is also provided including the fluid system, a first rotating and/or heat generating component and a second rotating and/or heat generating component. The first fluid chamber is configured for providing fluid to the first rotating and/or heat generating component, and the second fluid chamber is configured for providing fluid to the second rotating and/or heat generating component.

(17) A method of operating the fluid system is also provided including positioning the stopper in the first orientation; with the stopper in the first orientation, filling the first fluid chamber with fluid and pressuring the fluid in the first fluid chamber at a first pressure, and filling the second fluid chamber with fluid and pressuring the fluid in the second fluid chamber at a second pressure; and positioning the stopper in the second orientation to cause the fluid in the first fluid chamber and the fluid in the second fluid chamber to drain out of the drain hole at a same time.

(18) A method of constructing a fluid system for a motor vehicle drivetrain is also provided. The method includes forming a drain hole into a housing, and fluidically connecting to the drain hole a first fluid chamber for receiving fluid and a second fluid chamber for receiving fluid; and arranging a stopper in the drain hole, the stopper configured to prevent fluid from draining from the first fluid chamber and the second fluid chamber out of the drain hole in a first orientation of the stopper and configured to allow fluid to drain from the first fluid chamber and the second fluid chamber out of the drain hole in a second orientation of the stopper, the stopper configured to isolate the first fluid

chamber from the second fluid chamber in the first orientation of the stopper.

(19) In examples, the stopper is a drain plug, and arranging the stopper in the drain hole includes screwing the stopper into the drain hole.

(20) In examples, the drain plug includes an exterior section positioned outside of the drain hole in the first orientation and an interior section positioned inside of the drain hole in the first orientation, and the screwing the stopper into the drain holes including engaging the exterior section via a tool and rotating the drain plug to cause an external thread on the exterior section to advance along an internal thread of the drain hole.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present disclosure is described below by reference to the following drawings, in which:

(2) FIG. 1 shows a fluid system configured for use in the drivetrain of a motor vehicle with a stopper in a first orientation plugging a drain hole; and

(3) FIG. 2 shows the fluid system of FIG. 2 with the stopper in a second orientation removed from the drain hole.

DETAILED DESCRIPTION

(4) FIGS. 1 and 2 shows a fluid system **10** configured for use in the drivetrain of a motor vehicle. Fluid system **10** includes a housing **12** including drain hole **14**, a first fluid chamber **16** for receiving fluid, and a second fluid chamber **18** for receiving fluid. The drain hole **14** is coupled to the first fluid chamber **16** and the second fluid chamber **18** for draining fluid from both the first fluid chamber **16** and the second fluid chamber **18**.

(5) Fluid system **10** further includes a stopper **20** removably installable on the housing **12** and configured to prevent fluid from draining from the first fluid chamber **16** and the second fluid chamber **18** out of the drain hole **14** in a first orientation of the stopper **20**. Stopper **20** is also configured to allow fluid to drain from the first fluid chamber **16** and the second fluid chamber **18** out of the drain hole **14** in a second orientation of the stopper **20**. The first orientation of stopper **20** is shown in FIG. 1, and the second orientation of stopper **20** is shown in FIG. 2.

(6) In order to prevent pressure equalization of first fluid chamber **16** and second fluid chamber **18** during operation of the drivetrain, the stopper **20** is configured to isolate the first fluid chamber **16** from the second fluid chamber **18** in the first orientation of the stopper **20**. The stopper **20** is configured to prevent fluid from draining out of either the first fluid chamber **16** and the second fluid chamber **18** in the first orientation and to allow fluid to flow freely through the drain hole **14** in the second orientation. The stopper **20** also serves to isolate the first fluid chamber **16** from the second chamber when it is in the position where fluid flow is blocked.

(7) The drain hole **14**, the first fluid chamber **16** and the second fluid chamber **18** are configured to cause fluid from the first fluid chamber **16** and fluid from the second fluid chamber **18** to merge inside the drain hole **14** in the second orientation of the stopper **20**. Using a single drain hole **14** and a single stopper **20** allows the drainage of both of first fluid chamber **16** and second fluid chamber **18** simultaneously with as little effort and components as possible.

(8) The first fluid chamber **16** includes a first drain line **22** emptying into the drain hole **14** and the second fluid chamber **18** includes a second drain line **24** emptying into the drain hole **14**. Drain hole **14** is vertically aligned in the bottom of housing **12** and is centered on a vertically extending center axis **25**. Drain hole **14** is defined by a vertically extending side wall **14a** extending vertically upward from a bottom open end **14b** of drain hole **14**, through which the fluid from first fluid chamber **16** and second fluid chamber **18** drains, to a top open end **14c** of drain hole **14**. First drain line **22** extends horizontally from first fluid chamber **16** to drain hole **14**, and connects directly to side wall **14a**. Second drain line **24** extends horizontally from second fluid chamber **18** to drain

hole **14**, and connects directly to top open end **14c**. In other words, the first drain line **22** empties radially into the drain hole **14** with respect to the center axis **25**, and the second drain line **24** empties axially into the drain hole **14** with respect to the center axis **25**.

(9) The first fluid chamber **16** is configured for providing fluid to the first rotating and/or heat generating component **26** for lubricating and/or cooling the component **26**. The second fluid chamber **18** is configured for providing fluid to the second rotating and/or heat generating component **28** for lubricating and/or cooling the component **28**. In other words, each of chambers **16**, **18** provides fluid, which can for example be oil or another liquid, to a different component of the drive train. Components **26**, **28** can be components of an electric axle that drives the wheels of the vehicle. Components **26**, **28** can each be a gear set, an electric motor driving the wheels, power electronics or a battery module powering the electric motor.

(10) In the example shown in FIGS. **1** and **2**, stopper **20** is a drain plug including an interior section **30** positioned inside of the drain hole **14** in the first orientation and an exterior section **32** positioned outside of the drain hole **14** in the first orientation. Interior section **30** has a side wall **30a** having a shape that is complimentary to side wall **14a** of drain line **14** such that when stopper **20** is in the first orientation, side wall **30a** engages side wall **14a**. More specifically, the side wall **30a** of stopper **20** includes an external thread **30b** and the side wall **14a** of drain hole **14** includes an internal thread **14d**, and the external thread **30b** is screwed into the internal thread **14d** in the first orientation.

(11) Interior section **30** further includes an insertion end **30c** that is on the opposite side of side wall **30a** as exterior section **32**. When stopper **20** is inserted into drain hole **14**, insertion end **30c** is the first part of stopper **20** to enter into drain hole **14**.

(12) Two different surfaces of stopper **20** are used to block the outlets of the drain lines **22**, **24** when the stopper **20** is in the position where fluid flow out of drain hole **14** is blocked—i.e., in the first orientation of the stopper **20**. As shown in FIG. **1**, a first surface **30d** of interior section **30** blocks the outlet of the first drain line **22**, while a second surface **30e** of interior section **30** blocks the outlet of the second drain line **24**. This allows stopper **20** to block both of chambers **16**, **18** from draining while the stopper **20** is in the first orientation, and to isolate chambers **16**, **18** from each other. In the examples shown in FIGS. **1** and **2**, the first surface **30d** is formed on side wall **30a** of stopper **20** and second surface **30e** is formed on insertion end **30c** of stopper **20**.

(13) Exterior section **32** of stopper **20** is wider than interior section **30** of stopper **20**, which allows exterior section **32** to abut against an outer surface **12a** of housing **12**. In particular, exterior section **32** abuts against the outer surface **12a** of housing **12** via a seal **34** provided on a housing facing surface **32a** of exterior section **32**. In the example of FIGS. **1** and **2**, seal **34** is a ring and housing facing surface **32a**, which faces interior section **30**, is ring-shaped. Seal **34** is configured for sealing the first fluid chamber **16** and the second fluid chamber **18** from an environment outside of the housing **12** in the first orientation. The seal **34** is sandwiched between housing facing surface **32a** of the exterior section **32** and the housing **12** in the first orientation. Interior section **30** of stopper **20** is provided with a magnet **36** to collect all metal chips from the wear of the mechanical parts in contact with the fluid of fluid chambers **16**, **18**.

(14) Exterior section **32** of stopper **20** further includes, on a side of exterior section **32** facing away from interior section **30**, a head **32b** configured to be torqued by a tool to screw the stopper **20** into the drain hole **14** and into the first orientation, and to unscrew the stopper **20** from the drain hole **14** and into the second orientation. The stopper **20** is rotatable about vertically extending center axis **25** of drain hole **14** to screw the stopper **20** into the drain hole **14** and to unscrew the stopper **20** from the drain hole **14**.

(15) A method of operating the fluid system **10** includes positioning the stopper **20** in the first orientation within the drain hole **14** so the stopper **20** blocks the fluid from both of chambers **16**, **18** from flowing out of drain hole **14**, while also isolating chambers **16**, **18** from each other so that the pressure of the fluid in first fluid chamber **16** and the pressure of fluid in the second fluid chamber

18 are not equalized during operation of the drivetrain. Next, with the stopper **20** in the first orientation, the first fluid chamber **16** is filled with fluid and the fluid is pressurized to a first pressure, and the second fluid chamber **18** is filled with fluid and the fluid is pressurized to a second pressure different from the first pressure. The pressurization of the fluids in the first and second chambers **16, 18** can occur during operation of the motor vehicle drive train. After operating the motor vehicle drivetrain, the fluid can be drained from both of chambers **16, 18** for maintenance operations. This can advantageously be done in an efficient manner by removing stopper **20** from drain hole **14**. In particular, the method can include positioning the stopper **20** in the second orientation to cause the fluid in the first fluid chamber **16** and the fluid in the second fluid chamber **18** to drain out of the drain hole **14** at a same time.

(16) A method of constructing fluid system **10** includes forming drain hole **14** into housing **12**, and fluidically connecting first fluid chamber **16** and second fluid chamber **18** to drain hole **14**. The method also includes arranging stopper **20** in the drain hole **14** so the stopper **20** prevents fluid from draining from the first fluid chamber **16** and the second fluid chamber **18** out of the drain hole **14** in the first orientation of the stopper **20**. This can include screwing the stopper **20** into the drain hole **14** by rotating stopper **20** about center axis **25**. The screwing of the stopper **20** into the drain hole **14** includes engaging the exterior section **32** via a tool and rotating the stopper **20** to cause external thread **30b** on the interior section **30** to advance along internal thread **14d** of the drain hole **14**.

(17) In the preceding specification, the present disclosure has been described with reference to specific exemplary embodiments and examples thereof. It will, however, be evident that various modifications and changes may be made thereto without departing from the broader spirit and scope of present disclosure as set forth in the claims that follow. The specification and drawings are accordingly to be regarded in an illustrative manner rather than a restrictive sense.

LIST OF REFERENCE NUMBERS

(18) **10** fluid system **12** housing **12a** outer surface **14** drain hole **14a** side wall **14b** bottom open end **14c** top open end **14d** internal thread **16** first fluid chamber **18** second fluid chamber **20** stopper **22** first drain line **24** second drain line **25** center axis **26** components **28** components **30** interior section **30a** side wall **30** external thread **30c** insertion end **30d** first surface **30e** second surface **32** exterior section **32a** housing facing surface **32b** head **34** seal **36** magnet

Claims

1. A fluid system for a motor vehicle drivetrain comprising: a housing including drain hole, a first fluid chamber for receiving fluid, and a second fluid chamber for receiving fluid, the drain hole coupled to the first fluid chamber and the second fluid chamber for draining fluid from both the first fluid chamber and the second fluid chamber; and a stopper configured to prevent fluid from draining from the first fluid chamber and the second fluid chamber out of the drain hole in a first orientation of the stopper and configured to allow fluid to drain from the first fluid chamber and the second fluid chamber out of the drain hole in a second orientation of the stopper, the stopper configured for being insertable upward into the drain hole to isolate the first fluid chamber from the second fluid chamber in the first orientation of the stopper, wherein an outer circumference of the stopper blocks fluid flow from the first fluid chamber, the stopper including a magnet, wherein the magnet is in an interior section of the stopper along a center axis of the drain hole to collect metal chips from wear of mechanical parts in contact with the fluid of second chamber when the stopper is in the first orientation.

2. The fluid system as recited in claim 1 wherein the first fluid chamber includes a first drain line emptying into the drain hole and the second fluid chamber includes a second drain line emptying into the drain hole.

3. The fluid system as recited in claim 2 wherein the first drain line includes an outlet directly

connected to a side of the drain hole and the second drain line includes an outlet directly connected to a top of the drain hole.

4. The fluid system as recited in claim 2 wherein a first surface of the stopper blocks an outlet of the first drain line in the first orientation, and a second surface of the stopper blocks an outlet of the second drain line in the first orientation.

5. The fluid system as recited in claim 1 wherein the stopper includes a seal for sealing the first fluid chamber and the second fluid chamber from an environment outside of the housing.

6. The fluid system as recited in claim 1 wherein the stopper is a drain plug removably installable on the housing to plug the drain hole, a portion of the drain plug being installed in the drain hole in the first orientation, the drain plug being removed from the drain hole in the second orientation.

7. The fluid system as recited in claim 6 wherein the drain plug includes: an exterior section positioned outside of the drain hole in the first orientation; and the interior section positioned inside of the drain hole in the first orientation.

8. The fluid system as recited in claim 7 wherein the first fluid chamber includes a first drain line emptying into the drain hole and the second fluid chamber includes a second drain line emptying into the drain hole, wherein a first surface of the interior section of the drain plug blocks an outlet of the first drain line in the first orientation, and a second surface of the interior section of the drain plug blocks an outlet of the second drain line in the first orientation.

9. The fluid system as recited in claim 7 wherein the interior section of the drain plug includes an external thread and the drain hole includes an internal thread, the external thread being screwed into the internal thread in the first orientation.

10. The fluid system as recited in claim 9 wherein the exterior section of the drain plug includes a head configured to be torqued by a tool to screw the drain plug into the drain hole and into the first orientation, and to unscrew the drain plug from the drain hole and into the second orientation.

11. The fluid system as recited in claim 9 wherein the drain plug is rotatable about the center axis of the drain hole to screw the drain plug into the drain hole and to unscrew the drain plug from the drain hole, the first fluid chamber including a first drain line emptying radially into the drain hole with respect to the center axis, the second fluid chamber including a second drain line emptying axially into the drain hole with respect to the center axis.

12. The fluid system as recited in claim 7 wherein the drain plug includes a seal for sealing the first fluid chamber and the second fluid chamber from an environment outside of the housing in the first orientation, the seal being sandwiched between the exterior section and the housing in the first orientation.

13. The fluid system as recited in claim 1 wherein the drain hole, the first fluid chamber and the second fluid chamber are configured to cause fluid from the first fluid chamber and fluid from the second fluid chamber to merge inside the drain hole in the second orientation of the stopper.

14. A motor vehicle drivetrain comprising: the fluid system as recited in claim 1; a first rotating and/or heat generating component, the first fluid chamber configured for providing fluid to the first rotating and/or heat generating component; and a second rotating and/or heat generating component, the second fluid chamber configured for providing fluid to the second rotating and/or heat generating component.

15. A method of operating the fluid system as recited in claim 1, the method comprising: positioning the stopper in the first orientation; with the stopper in the first orientation, filling the first fluid chamber with fluid and pressuring the fluid in the first fluid chamber at a first pressure, and filling the second fluid chamber with fluid and pressuring the fluid in the second fluid chamber at a second pressure; and positioning the stopper in the second orientation to cause the fluid in the first fluid chamber and the fluid in the second fluid chamber to drain downward out of the drain hole at a same time.

16. A method of constructing a fluid system for a motor vehicle drivetrain, the method comprising: forming a drain hole into a housing, and fluidically connecting to the drain hole a first fluid

chamber for receiving fluid and a second fluid chamber for receiving fluid; and arranging a stopper in the drain hole, the stopper configured to prevent fluid from draining from the first fluid chamber and the second fluid chamber out of the drain hole in a first orientation of the stopper and configured to allow fluid to drain from the first fluid chamber and the second fluid chamber out of the drain hole in a second orientation of the stopper, the stopper configured for being insertable upward into the drain hole to isolate the first fluid chamber from the second fluid chamber in the first orientation of the stopper, wherein an outer circumference of the stopper blocks fluid flow from the first fluid chamber, the stopper including a magnet, wherein the magnet is in an interior section of the stopper along a center axis of the drain hole to collect metal chips from wear of mechanical parts in contact with the fluid of second chamber when the stopper is in the first orientation.

17. The method as recited in claim 16 wherein the stopper is a drain plug, and arranging the stopper in the drain hole includes screwing the stopper into the drain hole.

18. The method as recited in claim 17 wherein the drain plug includes: an exterior section positioned outside of the drain hole in the first orientation; and an interior section positioned inside of the drain hole in the first orientation, the screwing the stopper into the drain hole including engaging the exterior section via a tool and rotating the drain plug to cause an external thread on the interior section to advance along an internal thread of the drain hole.

19. The fluid system as recited in claim 1 wherein the housing is configured so that upon removal of the stopper from the drain hole, fluid from the second fluid chamber flows downward into the drain hole and then downward out of the drain hole.

20. The fluid system as recited in claim 19 wherein the housing is configured so that upon removal of the stopper from the drain hole, fluid from the first fluid chamber flows horizontally into the drain hole and then downward out of the drain hole.
